Final project manuscript

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete research-case evidence record.

Preserve failures, grouping and missing outcomes consistently across abstract, tables, figures and artifacts.

Keep one habit: Recall → W1–W3/W5 → independent case → project evidence → feedback and delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.
- Without opening the old lesson, explain one key idea from Independent reproduction check. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Writing results and limitations. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Negative results and claim revision. Give an example and a mistake that the idea helps you avoid.

Independent reproduction check

> **My remembered idea, example and correction**

Writing results and limitations

> **My remembered idea, example and correction**

Negative results and claim revision

> **My remembered idea, example and correction**

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A final manuscript aligns question, methods, results, limitations and reproducibility artifacts. Main raw result eighteen passes out of twenty evaluated is 90%, not the abstract's 100%; restore the two failures omitted from the figure and link every view to the correct record/version. Separate picture has A 7/10 = 70%, B 8/10 = 80% on indoor photos, a one-photo or ten-point descriptive gain.

  Ten photos come from five plants with two views each; report that grouping rather than ten independent plants. Five source IDs alone do not establish sampling independence. Case-level and plant-level outcomes are not supplied, so no invented transitions or plant-level rate are justified. Indoor coverage does not establish outdoor robustness, population superiority or significance.

  Keep exploratory versions and unassessed outcomes visible. Missing artifact IDs remain unknown. A complete manuscript is a traceable account, not a guarantee that the design or interpretation is valid.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

- A final manuscript aligns question, methods, results, limitations and reproducibility artifacts. Main raw result eighteen passes out of twenty evaluated is 90%, not the abstract's 100%; restore the two failures omitted from the figure and link every view to the correct record/version. Separate picture has A 7/10 = 70%, B 8/10 = 80% on indoor photos, a one-photo or ten-point descriptive gain.

  Ten photos come from five plants with two views each; report that grouping rather than ten independent plants. Five source IDs alone do not establish sampling independence. Case-level and plant-level outcomes are not supplied, so no invented transitions or plant-level rate are justified. Indoor coverage does not establish outdoor robustness, population superiority or significance.

  Keep exploratory versions and unassessed outcomes visible. Missing artifact IDs remain unknown. A complete manuscript is a traceable account, not a guarantee that the design or interpretation is valid.

- Abstract says 100% success; results table shows 18/20, and two failures are omitted from the figure. Show the intermediate steps and give the conclusion with its scope.

Main manuscript location	Current inconsistency	Supported correction
Raw results table	18/20	Keep original twenty-case record
Abstract	100% success	18/20 = 90% on these evaluated cases
Figure	Two failures absent	Restore both failures and total twenty
Failure count	Hidden by headline/figure	Two failures explicitly reported
Run/artifact references	Need consistency	Identify the actual linked record; no fabricated IDs
Broader claim	Not established	Keep case/outcome scope attached

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A final manuscript aligns question, methods, results, limitations and reproducibility artifacts. Main raw result eighteen passes out of twenty evaluated is 90%, not the abstract’s 100%; restore the two failures omitted from the figure and link every view to the correct record/version. Separate picture has A 7/10 = 70%, B 8/10 = 80% on indoor photos, a one-photo or ten-point descriptive gain.

  Ten photos come from five plants with two views each; report that grouping rather than ten independent plants. Five source IDs alone do not establish sampling independence. Case-level and plant-level outcomes are not supplied, so no invented transitions or plant-level rate are justified. Indoor coverage does not establish outdoor robustness, population superiority or significance.

  Keep exploratory versions and unassessed outcomes visible. Missing artifact IDs remain unknown. A complete manuscript is a traceable account, not a guarantee that the design or interpretation is valid.

- A report omits this qualification: A manuscript is ready only when its claims and artifacts align; formatting alone cannot certify **evidence**.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A final manuscript aligns question, methods, results, limitations and reproducibility artifacts. Main raw result eighteen passes out of twenty evaluated is 90%, not the abstract’s 100%; restore the two failures omitted from the figure and link every view to the correct record/version. Separate picture has A 7/10 = 70%, B 8/10 = 80% on indoor photos, a one-photo or ten-point descriptive gain.

  Ten photos come from five plants with two views each; report that grouping rather than ten independent plants. Five source IDs alone do not establish sampling independence. Case-level and plant-level outcomes are not supplied, so no invented transitions or plant-level rate are justified. Indoor coverage does not establish outdoor robustness, population superiority or significance.

  Keep exploratory versions and unassessed outcomes visible. Missing artifact IDs remain unknown. A complete manuscript is a traceable account, not a guarantee that the design or interpretation is valid.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Manuscript says 12 evaluated cases, raw record has 10 completed and 2 missing with 8 passes. Correct the summary.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- For Reproducible AI Research Study, use this week's independent task as a rehearsal for the final **evidence** pack: Manuscript says 12 evaluated cases, raw record has 10 completed and 2 missing with 8 passes. Correct the summary. Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	

Evidence	My record
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Final project manuscript. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Use the worked case for Final project manuscript. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Manuscript says 12 evaluated cases, raw record has 10 completed and 2 missing with 8 passes. Correct the summary.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

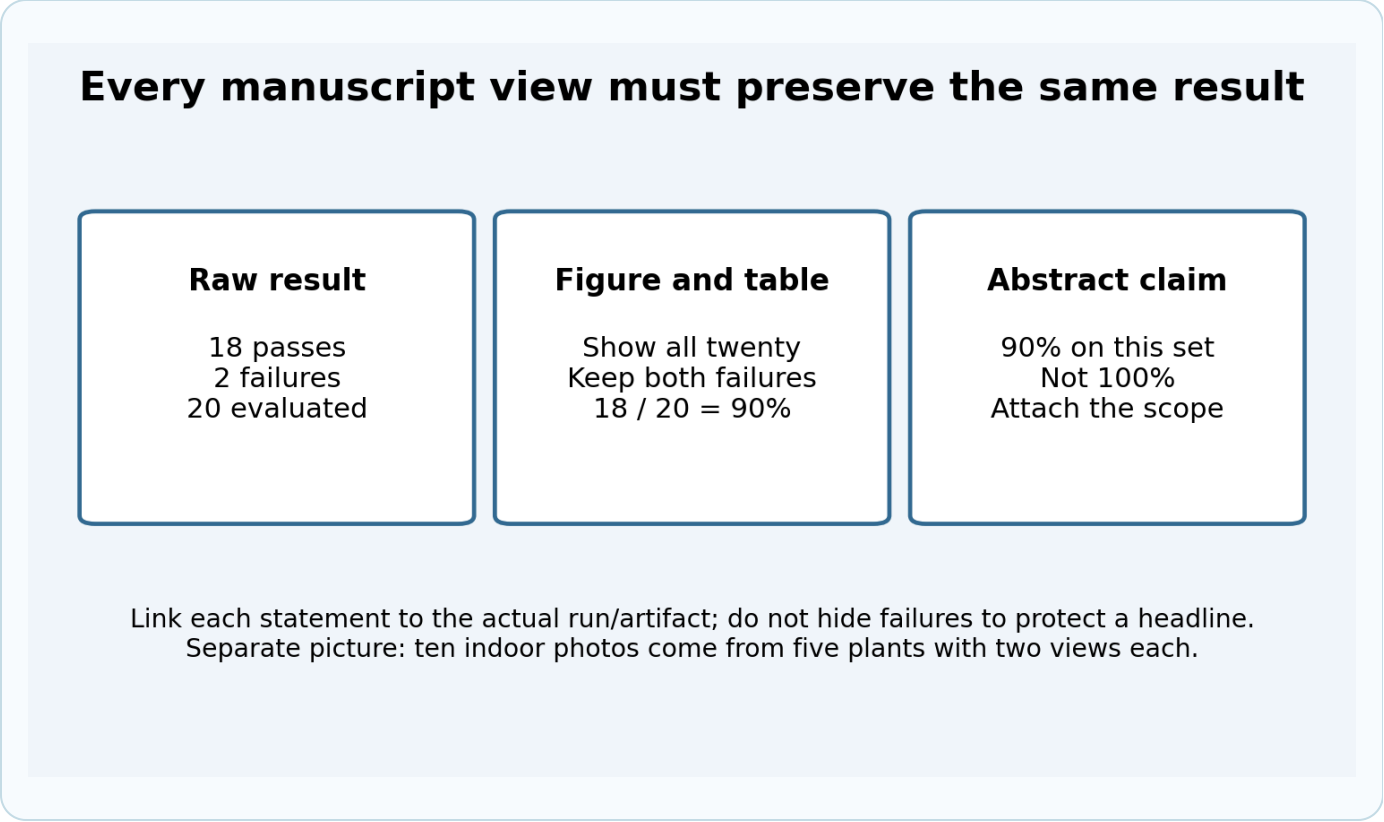
> Keep this task's inputs,rule and response together.

Complete task and evidence

- Build a manuscript from claim to record. A capstone manuscript compares two frozen pattern classifiers on ten reserved cases.
- A is correct on 7/10; B on 8/10. All cases are indoor, from five plants with two views each.
- Connect the manuscript's 7/10 and 8/10 scores to the five-plant grouping. Which abstract claim must be corrected if it says ten independent plants?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

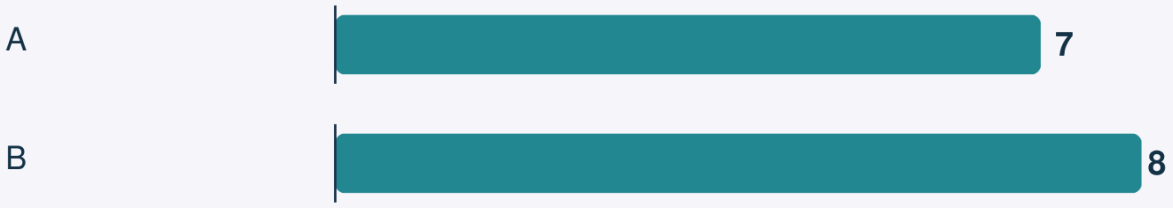
  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.



AI ACADEMY / AI-7/34

Indoor capstone photo outcomes

correct photos out of 10; five plants



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/34

Final project manuscript

Start with the supplied facts
Follow each arrow. Ask which fact allows the next step.

1. State bounded claim

B adds one correct photo on this indoor set.

2. Describe dependence

Ten photos come from five plants; report the grouping.

3. Present results

Show 70% and 80% with counts and case-level outcomes.

4. Preserve limitations

Do not claim outdoor robustness or ten independent plants.

CHANGE ONE THING
The abstract calls the study a test on ten independent plants.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ Reference · Project milestone

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Build a manuscript from claim to record. A capstone manuscript compares two frozen pattern classifiers on ten reserved cases.
- Which labelled step matches this detail? Do not claim outdoor robustness or ten independent plants.
- 1. Present results
- 2. Describe dependence
- 3. State bounded claim
- 4. Preserve limitations

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check after the first attempt

After your attempt, compare with the supported explanation: Preserve limitations Do not claim outdoor robustness or ten independent plants. A final manuscript aligns question, procedure, **evidence** and scope.

Every figure and conclusion should point back to the records that support it.

Changed-case question and optional hint

- Changed-case practice: The abstract calls the study a test on ten independent plants. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. Correct it to five plants with two views each.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Correct it to five plants with two views each. Five source plants have two views each; independence requires a justified sampling design.

Repair and retain the record

Review Independent reproduction check (ai-7/33). Goal: Explain the **mechanism** and terminology of independent reproduction check. Short practice: Locate the Q3 scoring disagreement between 5/6 and 4/6.

What should the teams inspect before choosing a score, and what limitation follows if the saved Q3 file is missing?. Return to this week and explain what you changed.

Review Writing results and limitations (ai-7/31). Goal: Explain the **mechanism** and terminology of writing results and limitations. Short practice: Compute the supported-answer rates for 9/12 and 10/12.

What actual outcome was measured, and why is a claim that every student becomes smarter outside this **evidence**?. Return to this week and explain what you changed.

Review Negative results and claim revision (ai-7/28). Goal: Explain the **mechanism** and terminology of negative results and claim revision. Short practice: Does a 6/8 tie support the **prediction** that B scores higher?

Explain why one newly selected favourable demonstration does not replace the original result or prove equivalence.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

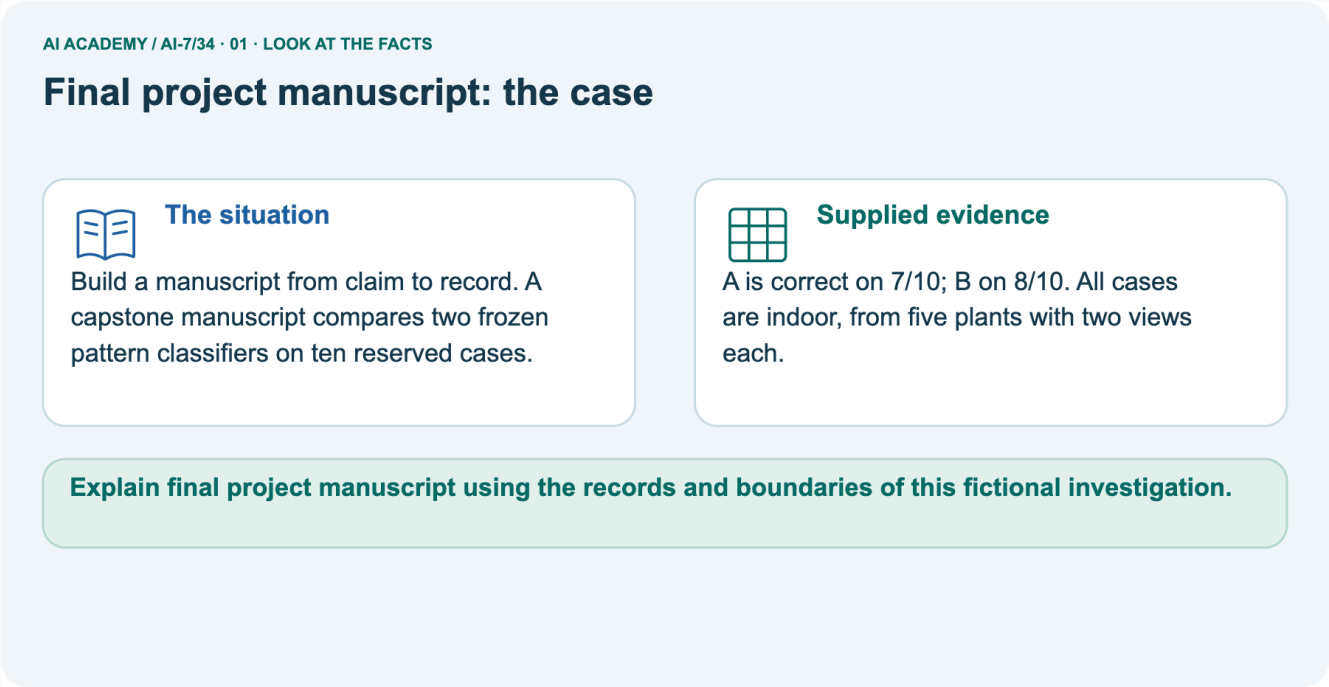
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Final project manuscript: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

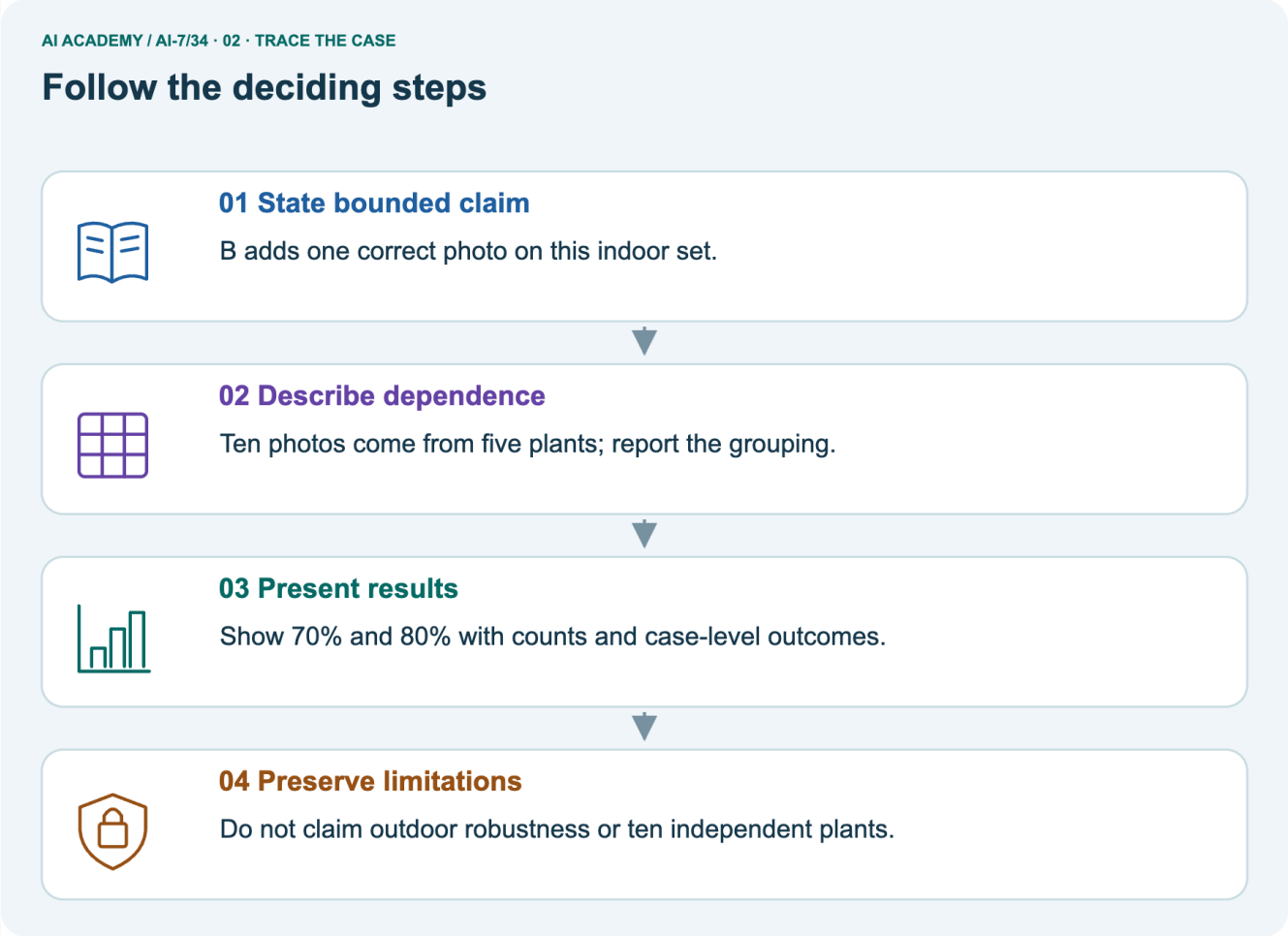
Build a manuscript from claim to record. A capstone manuscript compares two frozen pattern classifiers on ten reserved cases. Supplied facts: A is correct on 7/10; B on 8/10.

All cases are indoor, from five plants with two views each.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



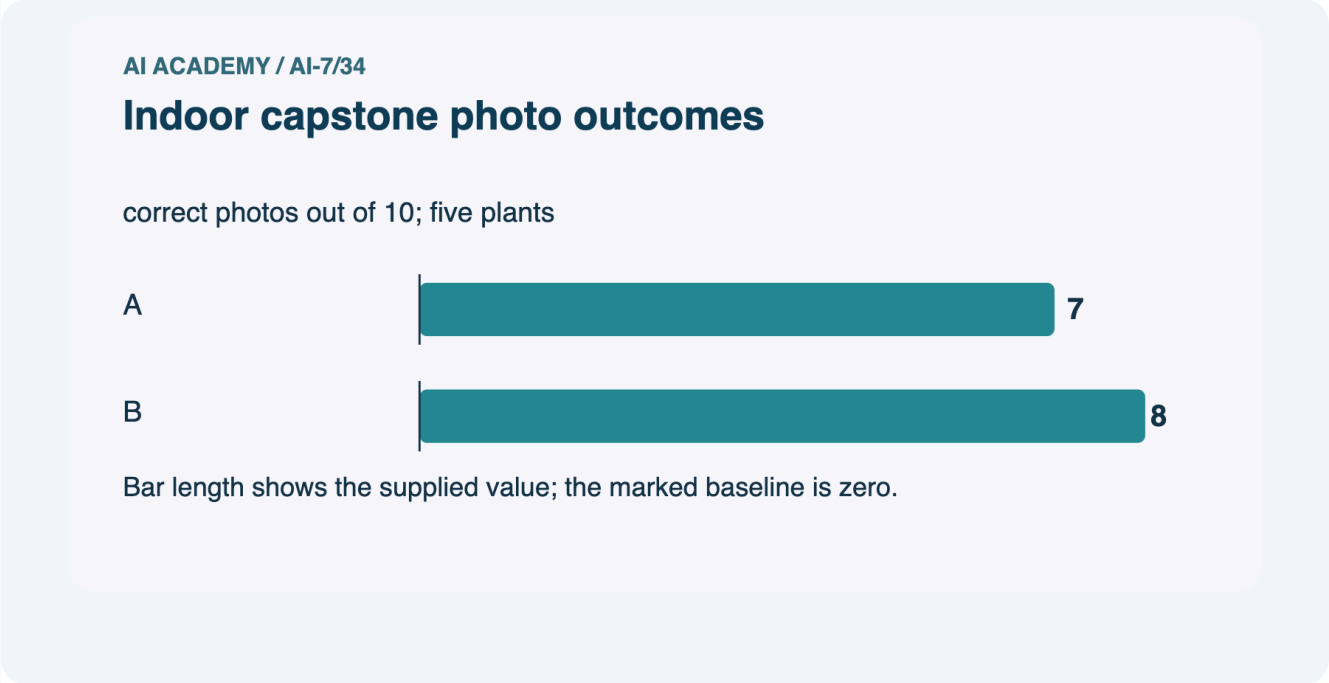
Follow the labelled picture with the matching explanation. Open the vector diagram

State bounded claim: B adds one correct photo on this indoor set. → Describe dependence: Ten photos come from five plants; report the grouping. → Present results: Show **70%** and **80%** with counts and case-level outcomes. → Preserve limitations: Do not claim outdoor robustness or ten independent plants.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Indoor capstone photo outcomes



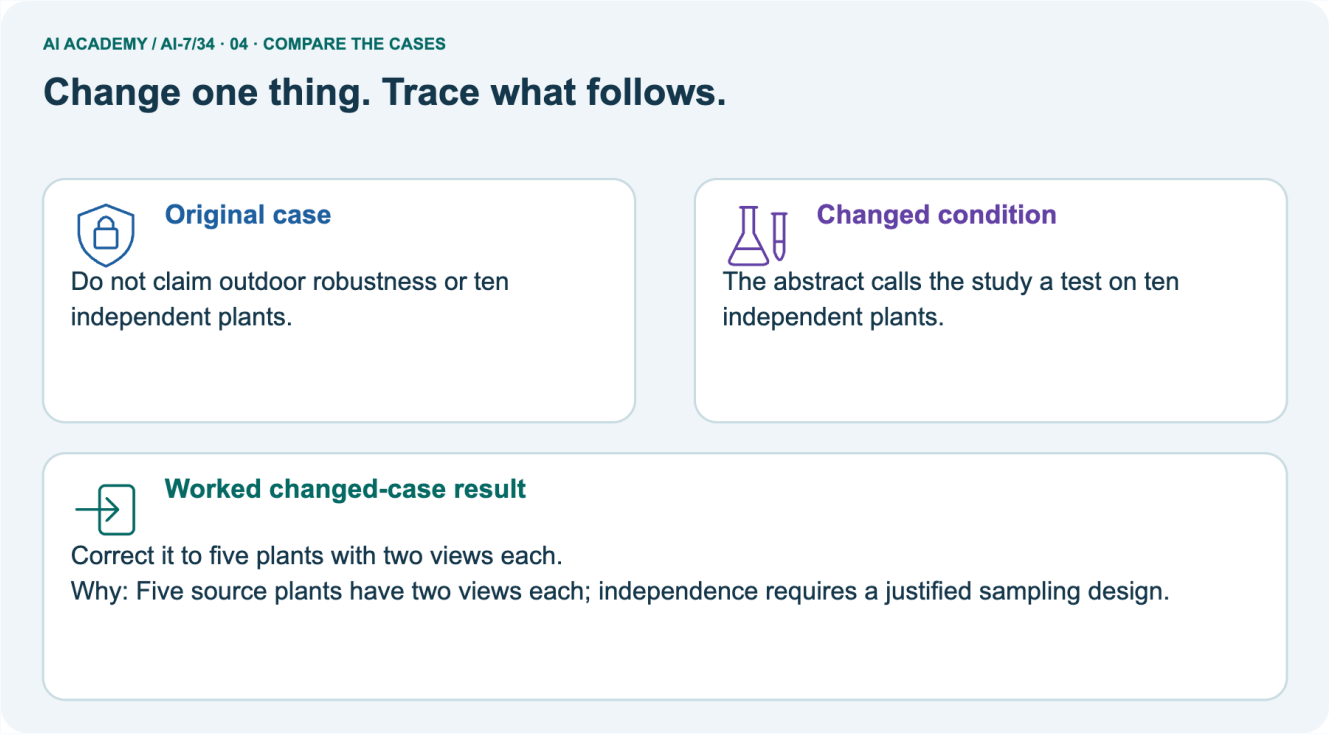
Follow the labelled picture with the matching explanation. Open the vector diagram

Indoor capstone photo outcomes. Units: correct photos out of 10; five plants. A: 7. B: 8.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



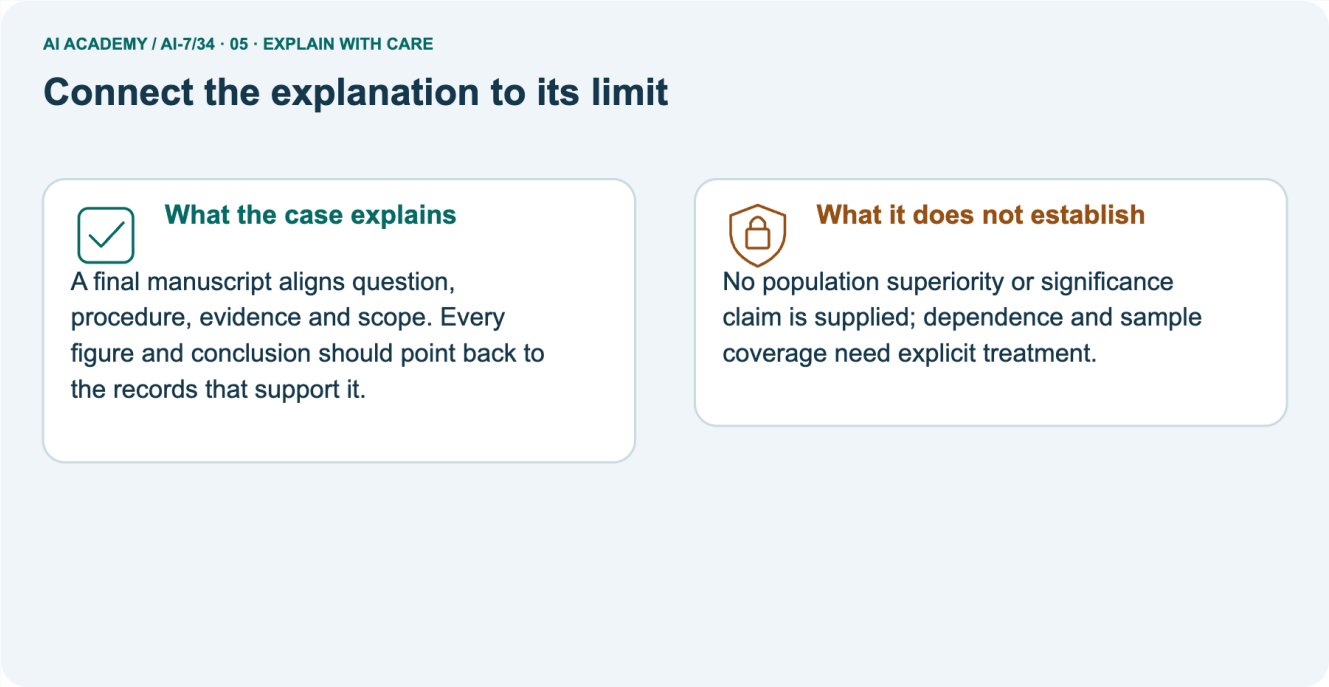
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Do not claim outdoor robustness or ten independent plants. Changed condition: The abstract calls the study a test on ten independent plants. Expected result: Correct it to five plants with two views each. Reason: Five source plants have two views each; independence requires a justified sampling design.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

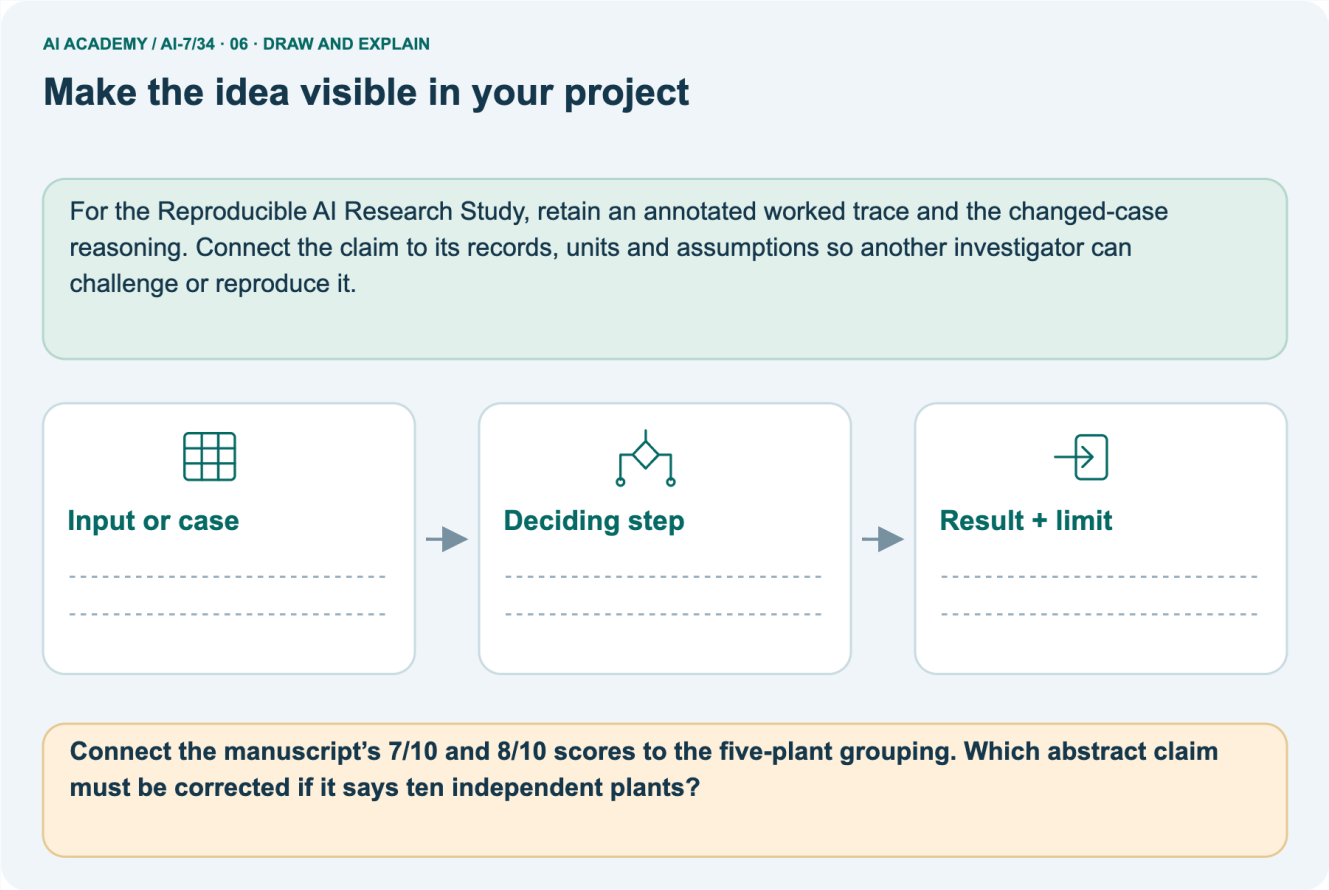
A final manuscript aligns question, procedure, **evidence** and scope. Every figure and conclusion should point back to the records that support it.

Limit: No population superiority or significance claim is supplied; dependence and sample coverage need explicit treatment.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Connect the manuscript's 7/10 and 8/10 scores to the five-plant grouping. Which abstract claim must be corrected if it says ten independent plants?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Final project manuscript: the case

02. Follow the deciding steps

03. Indoor capstone photo outcomes

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.